

The plant-system service desk

Teacher guide and worked answers

Explain how roots, stem, leaves and flowers contribute to the life of a flowering plant.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Summer | C31

Describe the job of each part of a plant.

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Identify roots, stem, leaves and flower.

Plants require water.

Success evidence

Match each main part to a useful job.

Explain one connection between parts.

Distinguish mineral uptake from making sugars.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Practical care

Paper and screen route requires no biological handling.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can one plant part do every job?
2	Retrieval	2-5	Name the part before its function
3	I do	5-12	I connect entry to transport

Slide	Stage	Minutes	Focus
4	I do	Within stage	I distinguish materials from sugar-making
5	I do	Within stage	I give flowers a biological function
6	We do	12-16	The service desk sorts the jobs
7	We do	16-20	A repair must address the missing input
8	Hinge	20-23	Which account explains plant nutrition?
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can one plant part do every job?

Do not anthropomorphise a service desk into intention: “jobs” means biological functions.

2 Retrieval Name the part before its function

1 roots. 2 stem. 3 flowers take part in seed formation. Accept an emerging answer without requiring pollination detail.

3 I do I connect entry to transport

Think aloud: “Water enters through the roots, but the leaves are higher up. I need a connecting pathway. Naming roots alone does not explain arrival at leaves.” Roots also anchor; arrows show selected water movement.

4 I do I distinguish materials from sugar-making

Shares I do time. Think aloud: “I see water arriving, but sugar-making also needs light and carbon dioxide from air. Soil is not a meal the roots eat.” Oxygen is also produced; the simplified diagram focuses on the inputs and sugar output.

5 I do I give flowers a biological function

Shares I do time. Think aloud: “A bright colour may help attract some pollinators, but it is not the whole function. I connect the flower with forming seeds.” Not every flower succeeds; not all flowers are bright or insect-pollinated.

6 We do The service desk sorts the jobs

W1: roots absorb/anchor; stem supports/transport; green leaves make sugars; flowers reproduction/seeds. Roles: explainer, evidence checker, connector. Rotate once.

7 We do A repair must address the missing input

W2: E needs suitable light reaching leaves; more minerals alone cannot replace light. F needs transport through the stem. Never damage or deliberately shade a class plant to act out a fault.

Reveal: A connected system needs suitable inputs and working pathways.

8 Hinge Which account explains plant nutrition?

Correct first option. If soil-food appears, revisit leafjob alongside rootjob.

A. Roots absorb water and minerals; green leaves make sugars using light. - This distinguishes uptake from sugar-making.

B. Roots eat soil, so leaves have no role. - Minerals and soil are not the plant's ready-made sugars.

C. Flowers pump all water from the soil. - Roots take in water, which travels through stem pathways.

9 You do Choose your science route

Set ONE route. Use the same objective across routes. A successful connection includes an entry point and transport, or a photosynthesis input and sugar-making. Do not demand the full photosynthesis equation.

10 You do Make your own evidence record

Use the same objective across routes. A successful connection includes an entry point and transport, or a photosynthesis input and sugar-making. Do not demand the full photosynthesis equation.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Use the same objective across routes. A successful connection includes an entry point and transport, or a photosynthesis input and sugar-making. Do not demand the full photosynthesis equation.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Parts have different functions and work together; water uptake is different from making sugars.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Parts have different functions and work together; water uptake is different from making sugars.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 roots. 2 stem. 3 flowers through reproduction and seed formation.

We do W1-W2

W1 roots take in water and stem carries it towards leaves; or stem supports leaves that receive light. Water entry roots; water transport stem; sugar-making green leaves; seed formation flowers. W2 light is missing; minerals cannot replace light energy.

Hinge

First option correct. The second treats soil as ready-made food; the third assigns root uptake to flowers.

S1-S5

S1 roots. S2 stem. S3 green leaves. S4 flowers. S5 needed substances taken in with water.

M1-M4

M1 roots absorb water, which travels through pathways in the stem to leaves. M2 light energy is needed to make sugars from water and carbon dioxide; water alone is insufficient. M3 supports leaves/flowers and transports substances. M4 flowers take part in reproduction and can lead to seeds.

T1-T4

T1 minerals do not replace the light energy needed for photosynthesis. T2 allow suitable light to reach leaves while maintaining other needs. T3 a connected stem pathway carries water beyond the roots. T4 no; seed development depends on successful reproductive processes and other suitable conditions.

Exit E1-E3

E1 roots. E2 green leaves and other photosynthetic green tissues; leaves sufficient for core. E3 reproduce/form seeds.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Secure core: four correct part-function links and one accurate connection. If a pupil says food comes from soil, contrast the labelled sugar output with root water/mineral uptake.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Roots eat soil as food.

Roots absorb water and dissolved minerals; green parts make sugars using light.

Check: Ask where sugars are made and where water enters.

The stem only holds the flower.

It supports parts and contains transport pathways.

Check: Ask how root-absorbed water reaches leaves.

Flowers exist only to look attractive.

They take part in reproduction and can lead to seed formation.

Check: Ask what a successfully fertilised flower can develop into.

Word	Meaning
absorb	Take in a substance, such as water.
mineral nutrient	A substance needed in small amounts for growth.
photosynthesis	Making sugars using light energy.
seed	A structure containing a young plant and food reserves.

Scientific references

[Department for Education: science programmes of study](#)

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

[Science and Plants for Schools: plant biology animations](#)

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.

[Science and Plants for Schools: water transport](#)

Checked 8 September 2026. Xylem transports water and dissolved minerals; evaporation from leaves contributes to pulling water upwards. Phloem sugar transport is distinct.